

Final Project Proposal

Team Name: (deep) fake it till you make it

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Project Description

For our project, we are looking to extend Gillian's midterm paper which investigated privacy preservation tools for AI image and text generation. One of her findings was that these tools are largely proactive (intended to be deployed by developers) rather than defensive (for use by individuals), and that the defensive tools that do exist are largely intended for detection rather than prevention. Of these, there is a larger focus on preventing deepfakes and photorealistic images than copyright infringement mitigation for artists, which is still being litigated in courts.

To explore this topic further, we are looking to apply the anti-AI image generation tool Glaze to combat AI art video generation [1]. Specifically, we aim to investigate what percent of video frames Glaze must be applied to in order to prevent AI generation of videos that copy certain artistic or animated styles.

Motivation

AI generated videos have become increasingly popular in recent years, including videos of fake actors, and AI-generated animation [2,3]. There are a number of tools that have been proposed to cloak real photos and videos from deepfakes [4,5], but few to prevent stylistic mimicry for art, and only for images [1]. Our goal is to apply a pre-existing tool for anti-AI style mimicry in images to videos. Users of this novel tool could apply it to their own animations and art videos to protect them from AI generation tools. Alongside further researching privacy preservation tools for AI imaging and text generation, we hope to empirically test our topic by experimenting with anti-AI image tools on the market.

Methodology

We will investigate what percent of frames must have Glaze applied in it in order to prevent artistic style copying via video. We will use some open-source animations online as well as create some simple ones to test our method against three text-to-video generation tools. We will evaluate its effectiveness through relation metrics of frames comparing the original and AI-generated videos trained on the Glazed video, similar to [4]. We will also apply some video similarity metrics specific to video and animation [6,7], and potentially surveys as well to compare against human metrics.

References

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